CHAPTER 2

Administration Basics

Welcome! In this chapter you'll get hands on with some basics of PostgreSQL operation and administration.

In the last chapter, you installed and configured the Rideshare Rails application connected to a database running on your local PostgreSQL server. If you skipped the Introduction and aren't familiar with Rideshare or don't have it installed, consider going back and reading that section. At the end you'll have a full development environment with the correct version of Ruby, Rails, and PostgreSQL installed, running on Mac OS. You'll be in a spot to run any Active Record code from `bin/rails console` for Rideshare, or run SQL statements for the Rideshare database.

PostgreSQL Administration is a broad subject and in this chapter you'll be getting an introduction to a variety of topics to help you branch out and go into greater depth.

There may be a lot of new terminology to learn. You'll work with meta-commands, psql, database activity observability, locks, database parameters, and extensions.

Most Rails developers don't write a lot of SQL directly to modify the database schema. Instead they write Active Record *Migrations* that are Ruby files that generate SQL DDL statements in order to evolve the database schema.

Since our goal is about operations and administrations knowledge, you'll begin to write more SQL modification statements and run them directly using a SQL client. You'll modify the database structure, populate data rows, and you'll write these statements with SQL.

Let's dive in using the Rideshare development database.